

Complex Analysis Homework 9

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4. By rearranging the series

$$\frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right],$$

show directly, without differentiation, that $\wp(z + \omega) = \wp(z)$ whenever $\omega \in \Lambda$.

[Hint: For R sufficiently large, note that $\wp(z) = \wp^R(z) + O(1/R)$, where

$$\wp^R(z) = z^{-2} + \sum_{0 < |\omega| < R} ((z + \omega)^{-2} - \omega^{-2}).$$

Next, observe that both $\wp^R(z+1) - \wp^R(z)$ and $\wp^R(z+\tau) - \wp^R(z)$ are $O\left(\sum_{R-c < |\omega| < R+c} |\omega|^{-2}\right) = O(1/R)$.]

证明 根据提示有

$$|\wp(z) - \wp^R(z)| = \left| \sum_{|\omega| \geq R} \frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right| \leq \sum_{|\omega| \geq R} \frac{c_1}{|\omega|^3} = O\left(\frac{1}{R}\right), \quad (1)$$

故而 $\wp(z) = \wp^R(z) + O(1/R)$. 而注意到

$$\wp^R(z) = \sum_{|\omega| < R} \frac{1}{(z + \omega)^2} + \sum_{0 < |\omega| < R} \frac{1}{\omega^2},$$

故而

$$\wp^R(z+1) - \wp^R(z) = \sum_{|\omega| < R} \frac{1}{(z+1+\omega)^2} - \frac{1}{(z+\omega)^2},$$

再考虑集合 $A = \{\omega \in \Lambda : |\omega| < R\}$ 和 $B = \{\omega \in \Lambda : |\omega - 1| < R\} = A + 1$, 令 $C = A \Delta B$, 我们取 c 足够大使得 $\sup_{\omega \in C} |\omega| \leq R + c$ 以及 $\inf_{\omega \in C} |\omega| \geq R - c$. 于是乎

$$\begin{aligned} |\wp^R(z+1) - \wp^R(z)| &\leq \sum_{\omega \in C} \left| \frac{1}{(z+\omega)^2} \right| \leq \sum_{R-c < |\omega| < R+c} \left| \frac{1}{(z+\omega)^2} \right| \\ &\leq c_2 \sum_{R-c < |\omega| < R+c} \frac{1}{|\omega|^2} = O\left(\frac{1}{R}\right). \end{aligned} \quad (2)$$

同理有

$$|\wp^R(z+\tau) - \wp^R(z)| = O\left(\frac{1}{R}\right). \quad (3)$$

综合 (1)(2)(3) 三式可知

$$\wp(z) = \wp^R(z + \omega) + O\left(\frac{1}{R}\right), \quad \forall \omega \in \Lambda,$$

令 $R \rightarrow \infty$ 即完成了证明.

6. Prove that $\wp''$ is a quadratic polynomial in $\wp$.

证明 $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$ 两边求导即得

$$\wp'' = 6\wp^2 - \frac{g_2}{2}.$$

3. Suppose Ω is a simply connected domain that excludes the three roots of the polynomial $4z^3 - g_2z - g_3$. For $\omega_0 \in \Omega$ and ω_0 fixed, define the function I on Ω by

$$I(\omega) = \int_{\omega_0}^{\omega} \frac{dz}{\sqrt{4z^3 - g_2z - g_3}}, \quad \omega \in \Omega.$$

Then the function I has an inverse given by $\wp(z + \alpha)$ for some constant α ; that is,

$$I(\wp(z + \alpha)) = z$$

for appropriate α .

[Hint: Prove that $(I(\wp(z + \alpha)))' = \pm 1$, and use the fact that $\wp$ is even.]

证明 $\wp$ 函数满足

$$(\wp'(z))^2 = 4\wp(z)^3 - g_2\wp(z) - g_3.$$

设

$$F(z) = I(\wp(z + \alpha)) = \int_{\omega_0}^{\wp(z + \alpha)} \frac{d\zeta}{\sqrt{4\zeta^3 - g_2\zeta - g_3}}.$$

由变上限积分求导法则与链式法则:

$$F'(z) = \frac{1}{\sqrt{4\wp(z + \alpha)^3 - g_2\wp(z + \alpha) - g_3}} \cdot \wp'(z + \alpha).$$

由微分方程

$$\sqrt{4\wp(w)^3 - g_2\wp(w) - g_3} = |\wp'(w)|,$$

因此

$$F'(z) = \frac{\wp'(z + \alpha)}{|\wp'(z + \alpha)|} = \pm 1.$$

由于 Ω 是单连通区域且避开三次多项式三根, 根式 $\sqrt{4z^3 - g_2z - g_3}$ 在 Ω 上取单值解析分支. 适当选取常数 α , 可使得

$$\wp'(z + \alpha) = \sqrt{4\wp(z + \alpha)^3 - g_2\wp(z + \alpha) - g_3},$$

从而 $F'(z) = 1$.

积分得 $F(z) = z + C$, 再调整 α 消去常数项 C , 即得

$$I(\wp(z + \alpha)) = z.$$

这说明 I 的逆函数可表示为 $\wp(z + \alpha)$.

4. Suppose τ is purely imaginary, say $\tau = it$ with $t > 0$. Consider the division of the complex plane into congruent rectangles obtained by considering the lines $x = n/2$, $y = tm/2$ as n and m range over the integers. (An example is the rectangle whose vertices are $0, 1/2, 1/2 + \tau/2$, and $\tau/2$.)

- (a) Show that $\wp$ is real-valued on all these lines, and hence on the boundaries of all these rectangles.
- (b) Prove that $\wp$ maps the interior of each rectangle conformally to the upper (or lower) half-plane.

证明 (a) 魏尔斯特拉斯椭圆函数满足共轭性质：若格满足 $\bar{\Lambda} = \Lambda$ ，则

$$\overline{\wp(z)} = \wp(\bar{z}).$$

由 $\tau = it$ ，有 $\bar{\tau} = -\tau$ ，故 $\bar{\Lambda} = \Lambda$ 。

对直线 $x = \frac{n}{2}$ ： $z = \frac{n}{2} + iy$ ，则

$$\bar{z} = \frac{n}{2} - iy, \quad \overline{\wp(z)} = \wp(\bar{z}) = \wp(z),$$

故 $\wp(z)$ 取实值。

对直线 $y = \frac{tm}{2}$ ： $z = x + \frac{m\tau}{2}$ ，同理

$$\overline{\wp(z)} = \wp(\bar{z}) = \wp(z),$$

$\wp(z)$ 也为实数。

因此 $\wp$ 在所有分割直线上取实值，从而在全体矩形边界上实值。

(b) $\wp^{-1}$ 是椭圆形积分，由 Schwarz-Christoffel 的理论，可知 $\wp^{-1}$ 将上（下）半平面映射到 $\frac{1}{4}$ 基础平行四边形，而由于是共形，因此 $\wp$ 将每个小矩形共形映射到上（下）半平面。